

ERASTEEL TECHNICAL DATASHEET: ASP 2078

Application Family: Cold work | Standards: High-Sulfur Machinable PM HSS

1. Grade Overview & Characteristics

ASP 2078 is a sulfurized high-alloy PM grade matching ASP 2060 base metallurgy with 0.23% added sulfur. This imparts exceptional machinability during complex tool fabrication while preserving 69 HRC working hardness.

2. Nominal Chemical Composition (wt. % range)

Element	Carbon (C)	Chromium (Cr)	Molybdenum (Mo)	Tungsten (W)	Vanadium (V)	Cobalt (Co)	Other
Range %	2.25 - 2.35	4.0 - 4.4	6.8 - 7.2	6.3 - 6.7	6.3 - 6.7	10.2 - 10.28	0.23% Sulfur for enhanced machinability

3. Mechanical Hardness & Property Profiles (Tempered state)

Metric	Value	Metric	Value (Rating 1-10)
Working Hardness Range	67.0 - 69.0 HRC	Machinability	6.5/10
Max Reachable Hardness	69.0 HRC	Wear Resistance	9.0/10
Annealed Delivery State	Max 280-350 HB	Toughness	5.0/10
Hot Hardness Retention	Secondary hardening	Hot Hardness	9.5/10
Heat Treatment Type	Vacuum / Salt bath	Grindability	5.5/10

4. Industrial Applications

Complex geometry cold blanking punches, gear shaper cutters, thread rolling dies, high-wear precision cold heading inserts.

5. Heat Treatment & Tempering Guidelines

Austenitizing at 1180-1210°C. Triple tempering at 560°C. Working hardness up to 69 HRC.